

THE UNIVERSITY OF DODOMA



COLLEGE OF INFORMATICS AND VIRTUAL EDUCATION
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CIVE DAY 2026 — PROJECT PRESENTATION
ACADEMIC YEAR: 2025/2026

PROJECT TITLE:
A DIGITAL PLATFORM FOR INTERNSHIP STUDENT TEACHER MANAGEMENT SYSTEM

MEMBERS

NO.	NAME	REGISTRATION NUMBER	PROGRAM
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1. DEPARTMENT & PROJECT IDENTIFICATION

DEPARTMENT	Computer Science and Engineering (CSE)
PROJECT TITLE	A Digital Platform for Internship Student Teacher Management System
TEAM MEMBERS	1. Haji Hamisi Huseni 2. Hamisi Selemani
SUPERVISOR	Mr. Barongo

2. INTRODUCTION

a. Background

In Tanzania, student teachers from universities and colleges are required to complete practical field training (Mafunzo ya Vitendo) in secondary and primary schools across the country. This involves placing students in schools, assigning supervisors (assessors), monitoring teaching activities through daily logbooks, and issuing completion certificates at the end of the internship period. Coordination spans multiple stakeholders: students, school assessors, head teachers and District Education Officers (DEOs). Traditionally, this entire process has been handled manually — through paper logbooks, physical assessment forms, and in-person coordination — leading to significant inefficiencies.

b. Problem Statement

The manual management of student teacher internships creates the following critical challenges:

- No centralized system to track hundreds of student teachers across regions and districts.
- Paper logbooks are prone to loss, forgery, and inconsistency in record-keeping.
- Assessors have no efficient digital tool to record evaluations and observations.
- DEOs and head teachers lack real-time visibility into student progress.
- Certificate issuance is delayed due to multi-level manual approval processes.
- Communication between university admin, schools, and students is slow and unreliable.

c. Main Objective

To design and develop a web-based Internship Management System (IMS) that fully digitizes and automates the field practice workflow — from student registration and school placement to logbook management, assessor evaluation, and certificate issuance — providing real-time monitoring and transparency for all stakeholders.

Specific Objectives:

- Automate student registration, school selection, and subject assignment.
- Enable digital logbook submission with assessor feedback and remarks.
- Provide assessors with tools to evaluate and grade student performance.
- Give DEOs and head teachers a dashboard for district-wide progress monitoring.
- Generate tamper-proof, digitally-signed completion certificates and official letters.
- Integrate AI tools to assist students in creating Schemes of Work and Lesson Plans.

3. THE CORE INNOVATION

a. System Architecture Diagram

[WEB BROWSER — HTTPS]				
Student Teacher Assessor Admin DEO Head Teacher				
▼ Django Web Application (Python 3.12 / Django 5.2) ▼				
AUTH MODULE Login / Register Password Reset	STUDENT MODULE School Selection Logbook / Reports	ASSESSOR MODULE Evaluations Remarks / Grades	BOARD / DEO Dashboard Approvals	ADMIN MODULE User Mgmt Assignments
AI TOOLS Scheme of Work Lesson Plans	PDF ENGINE Certificates Logbook PDFs	SMS / EMAIL Africa's Talking Gmail SMTP	CACHING Redis Session Store	REST APIs AJAX Endpoints JSON Responses
▼ Data Layer ▼				
PostgreSQL 16 Database (Schools / Students / Assessments / Logbooks)		Docker Container (Web + DB Isolated)	External Services • OpenAI / Google Gemini (AI) • Gmail SMTP (Email) • Africa's Talking (SMS)	

b. Methodology

Programming Language	Python 3.12
Web Framework	Django 5.2 (MVT Architecture)
Database	PostgreSQL 16 (via Docker)
AI Integration	OpenAI GPT-4 & Google Gemini — AI-powered Scheme of Work & Lesson Plan generation
PDF Generation	ReportLab — Certificates, Logbooks, Official Letters with watermarks & signatures
Containerization	Docker & Docker Compose — isolated, reproducible deployment
Frontend	HTML5, CSS3, JavaScript — Responsive multi-role web interface
Communication	Gmail SMTP (email credentials) ·